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Roll No. | | | | | | | | | | | | | **Total No. of Pages : 02**

Total No. of Questions : 09

B.Tech. (Electrical Engg.) (Sem.–7)

MICROCONTROLLER AND PLC

Subject Code : BTEE-702B-18

M.Code : 90576

Date of Examination : 04-07-22

Time : 3 Hrs. **Max. Marks : 60**

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains FIVE questions carrying FIVE marks each and students have to attempt any FOUR questions.**
3. **SECTION-C contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.**

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SECTION-A

1. Write briefly :

a) Draw the ladder command used for a digital input. Label with suitable address.

b) Differentiate between microprocessor and microcontroller.

c) What is advantage of PLC and hard wired system?

d) What do you understand by multiplexed address / data bus?

e) Draw the schematic of ADC interface with 8051 microcontroller on port 1 and 2.

f) Explain briefly the memory mapped I/O interfacing.

g) Differentiate between direct and indirect addressing mode of 8051.

h) Draw the hard reset circuit of 8051 microcntroller.

i) Explain briefly bit addressable RAM of 8051.

j) Discuss briefly how strings are handled in 8051.

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SECTION-B

2. Explain the architecture of 8051 with the help of a suitable diagram.

3. Discuss the timer organization of 8051 microcontroller.

4. Explain the scheme used to interface 4×4 hex keypad with 8051 at port 1 with the help of a suitable schematic diagram.

5. Explain the UART organization of 8051 microcontroller.

6. Explain the addressing scheme used in a typical PLC.

SECTION-C

7. In a typical application it is required to interface 8255 programmable peripheral interface with 8051 microcontroller using indirect memory addressing mode. Draw the schematic diagram of interfacing. Write down the configuration word to use the PORT A, B and C of 8255 as simple output ports. Write an assembly language program to transfer 0AAH on port A of 8255.

8. Explain the concept of counter functions used in PLC. Differentiate the features of various counter functions available in PLC.

9. Explain the Timer 1 module of 8051 microcontroller in detail. Draw all the Special Function Registers (SFR) related to Timer 0. Explain the function of each bit of SFR's.

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NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.